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Code and Creed: The Construction of AI-Islamic Discourse in Singapore's Media Landscape

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ABSTRACT

This study examines how Singapore's mainstream media shapes public understanding of artificial intelligence (AI) in Islamic contexts through analysis of newspaper coverage from 1989 to 2024. Drawing on computational analysis of over 620,000 articles from The Straits Times and The Business Times, we develop the concept of “double mediation” to explain how media simultaneously shapes understanding of both technological and Islamic domains while mediating their relationship. Through text analysis of 2978 AI-Islamic sentences, we identify distinct patterns across four technological eras: pre-internet (1989–1995), web (1996–2005), social media (2006–2015), and AI/ML (2016–2024). The findings reveal three key dynamics: increasingly sophisticated forms of double mediation, emergence of hybrid authority structures combining technological and religious expertise, and complex temporal politics in constructing techno-religious futures. We demonstrate how media institutions actively construct new frameworks for understanding through “authority bridging practices” and “temporal harmonization,” rather than merely mediating between existing domains.

1 | Introduction

In February 2024, the Mufti of Singapore, Dr. Nazirudin Mohd Nasir, delivered a speech during which he addressed the challenges that artificial intelligence (AI) imposes on the Muslim society and its use for Islamic contexts (MUIS 2024; The Straits Times 2024). This development exemplifies how Islamic institutions increasingly grapple with AI not merely as a technological innovation but as a force that fundamentally reshapes Islamic practice, knowledge transmission, and religious authority. The intersection of AI and Islam represents a unique sphere where classical Islamic questions of human consciousness (*ruh*), divine determination (*qadar*), and human stewardship (*khilafah*) collide with contemporary technological capabilities and aspirations. This intersection becomes particularly fascinating in Singapore, where a highly digitalized society coexists with a vibrant Muslim

community that actively engages in both regional and global Islamic discourse.

The intersection of religion and technology is not limited to Islam; however, it has long been characterized by both tension and creative adaptation across diverse religious traditions. From debates about printing sacred texts to contemporary controversies over genetic engineering, religious communities have consistently grappled with how technological innovations reshape practices, beliefs, and authority structures. Technology is never religiously neutral; rather, it becomes embedded in webs of meaning that reflect theological commitments and social power. Noble's (1997) historical analysis reveals how Western technological development has been deeply intertwined with religious imaginaries of transcendence and human perfectibility, while Postman (1985) argues that media technologies fundamentally

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alter the contexts in which religious messages are received and understood. McLuhan's (1964) observation that "the medium is the message" becomes particularly salient when examining how digital platforms reshape religious authority and practice. These broader theoretical insights help situate the specific case of AI-Islamic discourse within long-standing patterns of religious engagement with technological change.

The relationship between technology and Islamic thought has long attracted scholarly attention, from classical discussions of innovation (*bid'ah*) to contemporary Islamic responses to AI (Bunt 2018, 2024; Campbell and Evolvi 2020; Barzilai-Nahon and Barzilai 2005). Islamic scholars and institutions have historically engaged with technological innovations through sophisticated theological and jurisprudential frameworks, considering both their practical implications and their deeper effects on Islamic practice and belief (Hamid 2016). However, these relationships are not formed in a vacuum; they are actively constructed, negotiated, and disseminated through media discourse, particularly in contexts like Singapore where Islamic discourse intersects with state narratives of technological progress (Kong and Woods 2016).

Media outlets play a crucial role in shaping public understanding of both technological developments and Islamic perspectives, effectively serving a "double mediator" in this complex relationship. This role becomes particularly significant in the context of AI, where media coverage oscillates between techno-utopian visions of enhanced human capabilities and concerns about maintaining (Islamic) ethical frameworks and values (Bunt 2024; Slama 2018a, 2018b). When these narratives intersect with Islamic frameworks, they create unique discursive spaces where technological and religious authorities compete, collaborate, and redefine their roles within Singapore's carefully managed religious landscape.

Singapore presents an ideal case study for examining these dynamics for several reasons. As a highly developed city-state that actively positions itself as both a "Smart Nation" and a key center of Islamic finance and education in Southeast Asia, Singapore offers unique insights into how Muslim communities negotiate between technological advancement and Islamic values. The country's experience is particularly relevant given its position at the intersection of global Islamic networks and technological innovation, with institutions like MUIS playing crucial roles in mediating between Islamic tradition and technological modernity (Slama and Barendregt 2018).

The country's media ecosystem, particularly The Straits Times and The Business Times, has extensively covered both technological developments and Islamic affairs, providing a rich empirical basis for analyzing how these domains intersect. These publications operate within Singapore's distinctive media environment, where professional journalism coexists with careful consideration of religious sensitivities and national development priorities (George 2012). This creates a unique space for examining how Islamic perspectives on AI are constructed and communicated in public discourse.

This study addresses three key questions: How does mainstream media's framing of AI-Islam relationships reflect and

shape broader societal negotiations between Islamic and technological authority? How do different technological eras (pre-internet, web, social media, AI/ML) correspond to different Islamic interpretations of technology? What role does media play in constructing "techno-religious futures," and how does this shape public understanding of both domains? Through an analysis of media coverage from 1989 to 2024, this research reveals how Islamic and technological discourse have co-evolved, competing and combining in ways that reshape both domains.

Methodologically, this study employs computational text analysis of over 620,000 newspaper articles, combining natural language processing (NLP) techniques with qualitative analysis informed by Islamic studies and science and technology studies. This mixed-methods approach, building on recent developments in Islamic digital humanities (Bunt 2018; Campbell 2010; Campbell and Evolvi 2020), allows for both macro-level pattern identification and micro-level narrative analysis, revealing how media discourse has shaped and been shaped by changing technological and Islamic landscapes.

The study makes several theoretical contributions. First, it develops the concept of "double mediation" to explain how media simultaneously shapes understanding of both technological and Islamic domains, extending existing theories of mediatization (Hjarvard 2011; Hepp 2013) to account for specifically Islamic contexts. Second, it advances understanding of how Islamic institutions and communities engage with technological change, contributing to discussions of sociotechnical imaginaries (Jasanoff and Kim 2015) by examining how Islamic frameworks influence technological imagination and vice versa. Finally, it contributes to discussions of temporal politics by examining how Islamic and technological temporalities interact and compete in media discourse, with particular attention to Islamic concepts of time and progress.

These insights have significant implications for understanding how Muslim societies negotiate between technological progress and Islamic values, particularly in diverse, multi-religious contexts like Singapore. As AI continues to reshape human experience and capabilities, understanding how media shapes its relationship with religion becomes increasingly crucial for religious leaders, technology developers, and policymakers alike. This understanding is particularly important given Singapore's position in both Southeast Asian Islam and global technological development.

The article proceeds as follows: After reviewing relevant literature on mediatization of religion, sociotechnical imaginaries, and temporal politics, we present our theoretical framework of double mediation. We then describe the Singapore context and our methodological approach, followed by three findings chapters. The first examines the evolution of AI-Islam discourse across four technological eras, the second analyzes networks of authority between religious and technological actors, and the third explores the temporal politics of techno-religious futures. We conclude by discussing theoretical implications for understanding the complex interplay between media, technology, and religion in contemporary societies.

2 | Theoretical Framework: Media, Technology, and Islamic Authority

2.1 | Theorizing Double Mediation in Islamic Contexts

The concept of double mediation emerges from the intersection of two theoretical traditions: mediatization theory's understanding of how media shapes social institutions and Islamic scholarly perspectives on knowledge transmission and authority. Although existing mediatization theory, particularly Hjarvard's (2011) work, establishes how media institutions increasingly influence religious practice and authority, it has not fully addressed the unique dynamics that emerge when media simultaneously mediates both technological and religious understanding. This gap becomes particularly significant in Islamic contexts, where traditional concepts of knowledge transmission (*ilm*) and religious authority (*wilayah*) interact with contemporary media and technological systems in distinctive ways. This builds on broader anthropological understandings of religious mediation in digital contexts (Eisenlohr 2017; Selby and Sayeed 2023).

Islamic scholarly traditions have long recognized the complex relationship between medium and message, particularly in the transmission of religious knowledge. Classical debates about the role of written versus oral transmission of hadith, for instance, reveal sophisticated understandings of how different forms of mediation shape religious knowledge and authority (Bunt 2018). Contemporary Islamic engagement with digital technologies extends these traditional concerns while raising new questions about authority, authenticity, and religious practice in mediated environments (Campbell and Evolvi 2020). However, the current technological context, particularly the emergence of AI, introduces new complexities that require theoretical innovation.

The proposed concept thus advances existing theories of religious authority in several crucial ways. Although Barzilai-Nahon and Barzilai (2005) demonstrate how digital technologies transform traditional religious authority structures, and Turner (2007) describes the fragmentation of religious authority in digital contexts, double mediation reveals more complex patterns of authority construction and legitimation. Unlike earlier frameworks that primarily examine how religious authority adapts to technological change, double mediation shows how media institutions actively construct new forms of authority that simultaneously draw on both technological and religious legitimacy. This theoretical innovation builds on but moves beyond Campbell's (2010) religious-social shaping of technology approach by revealing how media coverage creates hybrid authority structures that transcend simple adaptation or resistance. Furthermore, although Bunt's (2018) analysis of cyber-Islamic environments focuses on how digital platforms transform religious authority, double mediation demonstrates how mainstream media actively constructs new frameworks for understanding and legitimating different forms of expertise. This approach helps explain why contemporary religious authority in technological contexts often exhibits legitimacy hybridity and the simultaneous deployment of multiple forms of authorization that draw on both religious tradition and technological expertise.

The concept of double mediation provides a framework for understanding how media institutions simultaneously shape understanding of both technological and Islamic domains while mediating their relationship to each other. This process operates through three distinct but interrelated mechanisms. First, media institutions translate both technological and Islamic concepts into forms accessible to public understanding, often using each domain to explain the other. For instance, AI capabilities might be described using Islamic concepts of knowledge and consciousness, whereas Islamic practices might be reframed through technological metaphors of processing and optimization. Second, media coverage creates "cultures of mediatization" (Hepp 2013) that shape how both technological and Islamic actors understand and relate to each other. Finally, media institutions establish frameworks for evaluating and legitimizing different forms of authority, creating authority bridges between technological and Islamic domains.

This double mediation becomes particularly significant in the context of AI, where media coverage must navigate between traditional Islamic frameworks for understanding consciousness, free will, and human dignity and the novel capabilities and challenges presented by AI. As Bunt (2024) demonstrates, contemporary Muslim communities increasingly engage with AI not merely as a technological tool but as a force that raises fundamental questions about Islamic concepts of knowledge, authority, and human nature. Media institutions play a crucial role in shaping how these engagements are understood and negotiated.

The framework of double mediation, furthermore, extends existing theoretical approaches in several ways. First, it builds on Campbell's (2010) religious-social shaping of technology approach by examining how media institutions actively construct relationships between technological and religious understanding rather than merely reporting on them. Second, it advances Slama and Barendregt's (2018) analysis of digital Islam (see also Raya 2024; Bunt 2018) by focusing specifically on how media shapes both technological and religious authority simultaneously. Finally, it contributes to broader discussions of sociotechnical imaginaries (Jasanoff and Kim 2015) by examining how media constructs future visions that incorporate both technological and Islamic elements.

In the Singapore context, this double mediation takes on particular significance given the country's position at the intersection of global technological networks and Islamic discourse. Singapore's media institutions must navigate between state narratives of technological progress and Islamic perspectives on technological change, creating imaginative spaces (Eickelman and Anderson 2003) where different forms of authority and "technological understanding" can coexist and interact.

By technological understanding, we refer to the collectively shared frameworks through which societies comprehend, evaluate, and imagine technological capabilities and limitations. This extends beyond technical knowledge to encompass the social construction of technology, wherein the meaning and implications of technological artifacts are actively negotiated through public discourse (Bijker 1995). In the context of AI, technological understanding includes not only comprehension of computational processes but also collective interpretations of

what AI can and should do, its relationship to human intelligence, and its broader social implications. When media institutions shape technological understanding, they influence which aspects of technology become salient, which concerns are foregrounded, and which possibilities are imagined. This process of meaning-making is particularly consequential in Islamic contexts, where technological understanding must be reconciled with established religious frameworks for comprehending human nature, knowledge, and moral responsibility. The concept of double mediation thus captures how media discourse simultaneously constructs what AI means technologically and what it means religiously, while actively negotiating the relationship between these interpretive frameworks.

It should be noted, however, that although this study focuses specifically on Islamic contexts, the concept of double mediation has broader applicability across religious traditions engaging with technological change. Christian communities navigating biotechnology, Jewish responses to reproductive technologies, and Buddhist engagements with artificial consciousness all exhibit similar patterns of media-facilitated negotiation between technological and religious understanding. What varies across religious contexts is not the fundamental process of double mediation but rather the specific theological frameworks, authority structures, and temporal orientations that shape how this mediation unfolds. Islamic contexts present distinctive features, including particular emphasis on textual authority, established jurisprudential methodologies for engaging with novel phenomena, and specific theological concepts regarding human nature and divine knowledge that inform technological interpretation. However, the underlying dynamic wherein media institutions simultaneously shape understanding of both technological capabilities and religious meaning while mediating their relationship represents a broader pattern worthy of comparative investigation across diverse religious and cultural contexts.

2.2 | Sociotechnical Imaginaries and Islamic Futures

The concept of sociotechnical imaginaries, as developed by Jasanoff and Kim (2015), provides a crucial framework for understanding how societies envision and pursue technological futures. However, when applied to Islamic contexts, this framework requires significant adaptation to account for how Islamic concepts of progress, knowledge, and human development shape technological imagination. In Muslim societies, technological futures are not merely imagined through secular frameworks of progress but are actively shaped by Islamic concepts of human stewardship (*khilafah*), ethical development (*akhlaq*), and divine knowledge (*ilm*). This intersection becomes particularly significant in contexts like Singapore, where Islamic institutions actively engage with technological development while maintaining traditional religious frameworks.

Islamic engagement with technological futures operates through “techno-religious imaginaries”—collectively held visions of the future that blend technological capabilities with Islamic ethical and spiritual aspirations. As Meyer (2020) demonstrates, these imaginaries are not simply religious responses to technological change but active forces that shape how technologies are under-

stood, developed, and implemented. In the context of AI, these imaginaries become particularly complex as they must navigate between Islamic concepts of consciousness and intelligence and the novel capabilities presented by artificial systems.

Media institutions play a crucial role in constructing and maintaining these techno-religious imaginaries. Building on Stolow’s (2013) analysis of religion and technological mediation, we can observe how media coverage creates imaginative bridges between technological and Islamic domains. These bridges operate through several mechanisms. First, media institutions translate both technological possibilities and Islamic principles into forms that can be mutually understood and evaluated. Second, they create narrative frameworks that allow for the coexistence of technological innovation and Islamic values. Finally, they establish criteria for evaluating the religious legitimacy of technological developments.

The temporal dimension of these imaginaries requires particular attention. Islamic concepts of time and progress, which often combine linear and cyclical elements, interact with technological temporalities in complex ways. As Hartog (2015) notes, different ways of understanding time create distinct horizons of expectation and possibility. In Islamic contexts, these temporal frameworks are further shaped by religious concepts of divine destiny (*qadar*) and human responsibility. Media coverage must navigate between these different temporal orientations while constructing coherent narratives about technological futures.

Singapore’s position as both a technological hub and a place of Islamic discourse makes it particularly suitable for examining these dynamics. As Slama and Barendregt (2018) demonstrate in their analysis of digital Islam in Asia, Southeast Asian Muslim communities have developed distinctive approaches to technological modernity that blend Islamic principles with technological innovation. Singapore’s media institutions play a crucial role in this process, creating imaginative spaces where multiple authorities can interact.

The emergence of AI has intensified these dynamics by raising fundamental questions about consciousness, agency, and human uniqueness—questions that Islamic thought has long engaged with through theological and philosophical frameworks. As Bunt (2024) shows in his analysis of digital Muslim practices, contemporary Islamic engagement with AI often draws on classical theological concepts while developing new frameworks for understanding technological change. Media coverage of AI in Islamic contexts must therefore navigate between traditional religious frameworks and emerging technological possibilities while maintaining coherent narratives about future development.

These sociotechnical imaginaries also have significant implications for how authority is constructed and maintained in Muslim societies. Building on Campbell’s (2010) analysis of religious-social shaping of technology, we can observe how media institutions help establish authority frameworks that allow for the legitimization of both technological and religious expertise. These frameworks become particularly important in contexts where traditional religious authority intersects with technological expertise, creating new hybrid forms of authority that draw on both domains.

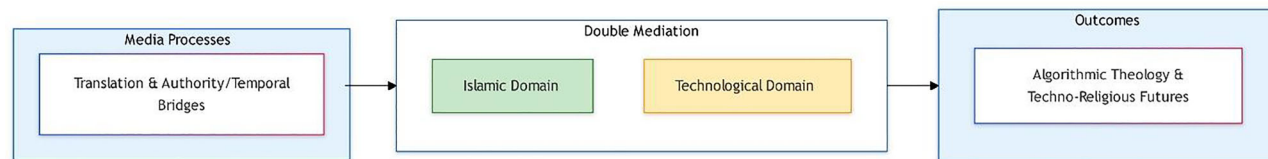


FIGURE 1 | Theoretical conceptualization of double mediation. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/jsr.70056)]

2.3 | Theoretical Synthesis: An Islamic Framework for Double Mediation

Drawing together the threads of mediatization theory, sociotechnical imaginaries, and Islamic perspectives on technology, we now develop an integrated theoretical framework for analyzing how media shapes the relationship between AI and Islam (Figure 1). This synthesis addresses three key theoretical innovations: the refinement of double mediation in Islamic contexts, the development of algorithmic theology, and a framework for analyzing temporal politics in techno-religious discourse. These elements combine to create an approach for understanding how media institutions simultaneously shape both technological and Islamic understanding while mediating their relationship.

The concept of double mediation, when applied specifically to Islamic contexts, reveals distinctive patterns of authority negotiation and knowledge transmission. Building on Hjarvard's (2011) mediatization theory while incorporating Islamic scholarly perspectives on knowledge (*ilm*) and authority (*wilayah*), we can identify three key mechanisms through which this double mediation operates. First, media institutions translate both technological and Islamic concepts into mutually intelligible frameworks. Second, they establish authority bridges that allow for the legitimation of both technological and religious expertise. Finally, they create narrative frameworks that enable the coexistence of technological innovation and Islamic values while acknowledging potential tensions between them.

This process of double mediation gives rise to algorithmic theology—new forms of religious understanding that emerge from the intersection of Islamic thought and computational systems. Although double mediation describes the broader process through which media institutions simultaneously shape technological and religious understanding, algorithmic theology represents the specific theological frameworks that emerge from this mediation process. As Bunt (2024) demonstrates, contemporary Islamic engagement with AI often draws on classical theological concepts while developing new frameworks for understanding technological change. This dialectical relationship between double mediation and algorithmic theology operates through three distinct mechanisms: first, through the recursive use of computational metaphors in religious discourse (where algorithmic processes help explain divine operations and vice versa); second, through the bidirectional application of Islamic ethical principles to AI development (where religious frameworks shape technological development while being simultaneously transformed by it); and third, through the emergence of new forms of religious authority that combine technological and Islamic expertise. Algorithmic theology thus represents not merely an outcome of double mediation but an active force that shapes how media institutions construct relationships between

technological and religious domains. This theoretical relationship helps explain why media coverage increasingly employs hybrid conceptual frameworks that blend computational and theological understanding, reflecting new forms of religious meaning-making in technological contexts.

The temporal dimension of this framework requires particular attention, especially given the distinctive ways that Islamic and technological temporalities interact in media discourse. Although double mediation operates as an overarching process of meaning-making between technological and religious domains, its temporal aspects manifest through temporal bridges—narrative frameworks that enable and actively construct the coexistence of different temporal orientations. Building on Hartog's (2015) analysis of temporal regimes while incorporating Islamic concepts of time and progress, we argue that these temporal bridges function as crucial mechanisms within the double mediation process, actively reconciling seemingly incompatible temporal frameworks. As El-Shamsy (2020) demonstrates, Islamic temporality operates through complex interplays between cyclical religious time and linear historical progression. Temporal bridges, therefore, serve not merely as passive connectors but as active sites of temporal negotiation where media institutions construct new hybrid temporalities that blend Islamic concepts of divine destiny (*qadar*) and human responsibility with technological narratives of progress and innovation. This theoretical relationship helps explain why media coverage of AI in Islamic contexts frequently exhibits temporal hybridity and the simultaneous deployment of multiple temporal frameworks that allow for both technological advancement and religious continuity. These bridges become particularly important in coverage of AI development, where media must navigate between traditional Islamic eschatology, contemporary technological determinism, and emerging forms of algorithmic temporality.

In the context of Singapore's media environment, this theoretical synthesis provides several analytical advantages. First, it allows us to understand how media institutions navigate between state narratives of technological progress and Islamic perspectives on technological change. Second, it helps explain the emergence of new forms of authority that combine technological and religious expertise. Finally, it provides a framework for analyzing how media constructs future visions that incorporate both technological and Islamic elements.

This theoretical framework advances existing scholarship in several ways. It extends Campbell's (2010) religious-social shaping of technology approach by examining how media actively constructs relationships between technological and religious understanding. It builds on Slama and Barendregt's (2018) analysis of digital Islam by focusing specifically on the role of media in shaping both technological and religious author-

ity. Additionally, it contributes to discussions of sociotechnical imaginaries (Jasanoff and Kim 2015) by examining how religious frameworks influence technological imagination and vice versa. Relatedly, the framework's application to AI coverage in Islamic contexts reveals several key dynamics. Media institutions not only report on technological developments but actively shape how these developments are understood within Islamic frameworks. They create mediating spaces (Meyer 2020) where different forms of authority and understanding can coexist and interact. Through this process, they help establish new forms of religious-technological authority that draw on both domains while maintaining distinctive characteristics of each.

3 | Context and Methodology

3.1 | Singapore Context

Singapore presents a unique case study for examining the intersection of AI and Islam due to its distinctive position as both a technological hub and a society with a significant Muslim minority. The city-state's commitment to becoming a "Smart Nation" through aggressive technological adoption coincides with its carefully maintained religious pluralism, where Muslims constitute approximately 14% of the population (Kong and Woods 2016). This intersection becomes particularly significant in the context of AI development, where Singapore's push for technological innovation frequently encounters Islamic ethical frameworks and considerations.

The Islamic landscape in Singapore is characterized by its institutional organization under the Islamic Religious Council of Singapore (MUIS), which plays a crucial role in mediating between technological advancement and Islamic principles. MUIS's recent engagement with AI-related issues, including the use of AI in religious contexts (Farouq 2023), reflects broader tensions between technological innovation and Islamic practice. This institutional framework operates within Singapore's broader technological landscape, which has evolved rapidly since the 1980s, with the government actively promoting digital transformation across all sectors of society, including religious institutions (Woods 2019).

Singapore's approach to both media regulation and Islamic-technological issues is governed by a regulatory framework that emphasizes social harmony and national development. The Administration of Muslim Law Act (AMLA) and the Maintenance of Religious Harmony Act (MRHA) provide guidelines for Islamic discourse in public spaces, including media coverage (Cohen and Tan 2015). Recent regulatory developments, such as the Protection from Online Falsehoods and Manipulation Act (POFMA) and MUIS engagement with AI usage in Islamic contexts (MUIS 2024; The Straits Times 2024; Farouq 2023), have added new dimensions to this regulatory landscape.

Singapore's media environment, similarly, is characterized by a high degree of government oversight combined with professional journalistic standards. The two primary newspapers analyzed in this study, The Straits Times and The Business Times, operate under Singapore Press Holdings (SPH) and represent the main English-language news sources covering both technological developments and Islamic affairs. These publications have

played a crucial role in mediating public understanding of both technological advancement and religious practice.

The coverage of Islamic issues in Singaporean media occurs within a carefully managed framework that emphasizes religious harmony while promoting technological progress. This creates a distinctive environment where media coverage of AI and Islam often aligns with broader national narratives about modernization and social cohesion, while remaining sensitive to religious sensibilities. The media's role as a mediator between technological and Islamic domains becomes particularly significant given Singapore's emphasis on managed public discourse around sensitive topics.

3.2 | Data Collection and (Pre-)processing

Following established approaches in computational religious studies (Campbell and Evolvi 2020), our corpus comprises articles published in The Straits Times and The Business Times between 1989 and 2024, focusing on coverage that intersects AI and Islamic themes. Article selection followed a rigorous multi-stage process using both computational and manual methods. The initial identification of relevant articles utilized a list of 744 AI-related terms and 679 Islamic terms, developed through consultation with Islamic scholars and technology experts. These terms were compiled in English and Malay to account for Singapore's multilingual Islamic discourse. The initial dataset consisted of 620,098 articles, which were systematically processed using methods adapted from recent studies in digital religion (Golan and Martini 2019). Following deduplication and filtering procedures developed for text analysis (Goldenstein and Poschmann 2019), the corpus was refined to 281,738 unique articles, of which 199,803 contained AI-related content and 25,211 specifically addressed both AI and Islamic contexts. Our selection criteria were designed to capture both explicit and implicit discussions of AI-Islam relationships, following methodological approaches established in recent studies of Islamic digital culture (Campbell and Evolvi 2020). Articles were included if they contained substantive discussion of both technological and Islamic themes, rather than merely incidental mentions.

We operationalized substantive discussion through multiple criteria applied systematically across the corpus. First, articles were required to contain at least three co-occurrences of AI-related and Islamic terms within a five-sentence window, ensuring meaningful engagement rather than coincidental mention. Second, articles needed to demonstrate semantic coherence, wherein AI and Islamic concepts were integrated into unified arguments or narratives rather than appearing in separate, unrelated sections (Grimmer and Stewart 2013). Finally, human validation was conducted on a random sample of articles to verify automated classification accuracy. Articles failing these criteria were classified as containing only incidental mentions and excluded from detailed analysis. This rigorous filtering process ensured that our final corpus of 2978 AI-Islamic sentences represented genuine engagement with techno-religious questions rather than spurious co-occurrence of terminology.

The temporal span of the corpus (1989–2024) captures the evolution of AI-Islam discourse across four distinct technological

eras: the pre-internet era (1989–1995), the web era (1996–2006), the social media era (2007–2016), and the AI/ML era (2017–2024). This periodization builds on established frameworks for studying technological change in religious contexts (Campbell 2010). The selection of these four eras reflects both technological infrastructure changes and demonstrable shifts in discourse patterns evident in our corpus. The pre-internet era (1989–1995) corresponds to the period before widespread public internet access. The web era (1996–2006) captures the rise of internet connectivity and digital information access, ending with the emergence of social media platforms that fundamentally altered information sharing practices. The social media era (2007–2016) begins with Facebook’s expansion beyond university networks in 2006 and Twitter’s launch in 2006, encompassing the period when social media became dominant platforms for religious discourse, as documented by Campbell and Garner (2016). The AI/ML era (2017–2024) marks the period when machine learning technologies moved from specialized research contexts to widespread public application, catalyzed by developments such as the widespread adoption of deep learning architectures and large language models.

The data processing pipeline employed a combination of established NLP techniques and custom-developed tools for Islamic text analysis. Raw text was purchased and extracted from SPH archive with specific attention to accurate rendering of Islamic terminology and proper names (Belinkov et al. 2019). The extracted text underwent systematic cleaning and standardization procedures, including removal of duplicates, standardization of formatting, and correction of errors. Text preprocessing included tokenization, removal of stop words, and lemmatization, with special attention paid to maintaining the integrity of Islamic terminology and technical vocabulary. The pipeline was developed using R and Python programming languages, incorporating specific modifications for handling Islamic texts and terms.

3.3 | Analytical Approach

The analysis employed a mixed-methods approach combining computational text analysis with qualitative interpretation, building on recent advances in Islamic digital humanities (Perrier and Salah 2024). Primary computational methods included named entity recognition (NER) for identifying key Islamic actors and institutions (Nadeau and Sekine 2007), topic modeling for identifying thematic patterns in Islamic discourse (Blei et al. 2003), sentiment analysis for examining emotional valence (Liu 2015), and word embeddings for analyzing semantic relationships between technological and Islamic concepts (Mikolov et al. 2013). Network analysis was employed to examine relationships between actors, institutions, and concepts within Singapore’s Islamic technological discourse. The analysis utilized both traditional social network analysis metrics and custom-developed measures to capture Islam-specific relationships (Golan and Martini 2019).

4 | Findings

The analysis of Singapore’s media coverage of AI in Islamic contexts between 1989 and 2024 reveals distinct patterns in

how technological and religious understanding have been co-constructed through media discourse. The findings demonstrate three key dimensions of this relationship: the evolution of AI-religion discourse across technological eras, the construction of authority networks, and the development of techno-religious future imaginaries. These dimensions reflect the complex ways in which media institutions have shaped both technological and religious understanding while mediating their relationship to each other. Through systematic analysis of media coverage, we can observe how double mediation operates in practice, how authority structures evolve, and how temporal frameworks are constructed and negotiated.

4.1 | Evolution of AI-Religion Discourse

The analysis of 2978 AI-Islamic sentences spanning from 1989 to 2024, visualized in Figure 2, reveals four distinct eras in how Singapore’s media has mediated the relationship between AI and Islamic thought (Hoffmann and Miller 1997). This evolution reflects not only technological advancement but also fundamental shifts in how Islamic frameworks engage with and interpret AI.

The pre-internet era (1989–1995) was characterized by “foundational mediation,” where media coverage primarily focused on establishing basic frameworks for understanding the relationship between Islamic principles and computational technology. The visualization shows that during this period, “ethics, policy, and governance” as “general technology” frames dominated, accounting for approximately 70% of coverage combined. This quantitative dominance of ethical and governance frames reflects “black boxing” of technology (Latour 1987), wherein computational systems were treated as settled objects requiring moral evaluation rather than as contested sociotechnical assemblages. The patterns thus demonstrate how early double mediation operated primarily through ethical translation rather than theological integration, establishing foundational authority structures that would later be complicated by more sophisticated engagement.

The web era (1996–2005) marked a significant shift toward “integrative mediation,” reflected in coverage that showed increasing sophistication in relating Islamic concepts to emerging digital capabilities. The visualization indicates a notable rise in “innovation and future trends” framing, growing from approximately 15%–30% of coverage. This increase in innovation framing provides quantitative evidence of the shift from reactive ethical evaluation to proactive future-making, wherein media discourse began actively constructing techno-religious imaginaries. This computational pattern aligns with theoretical expectations that web-era infrastructure would enable new forms of authority negotiation, as the decreased barrier to information circulation challenged traditional gatekeeping mechanisms while creating new mediating roles for journalistic institutions. This period also saw the emergence of cyber-Islamic environments (Bunt 2018), with media coverage increasingly focusing on how Islamic practices could be transmitted and transformed through digital means.

The social media era (2006–2015) witnessed the emergence of “networked mediation,” characterized by more complex engage-

Evolution of AI-Islamic Framing

Showing Different Technological Eras

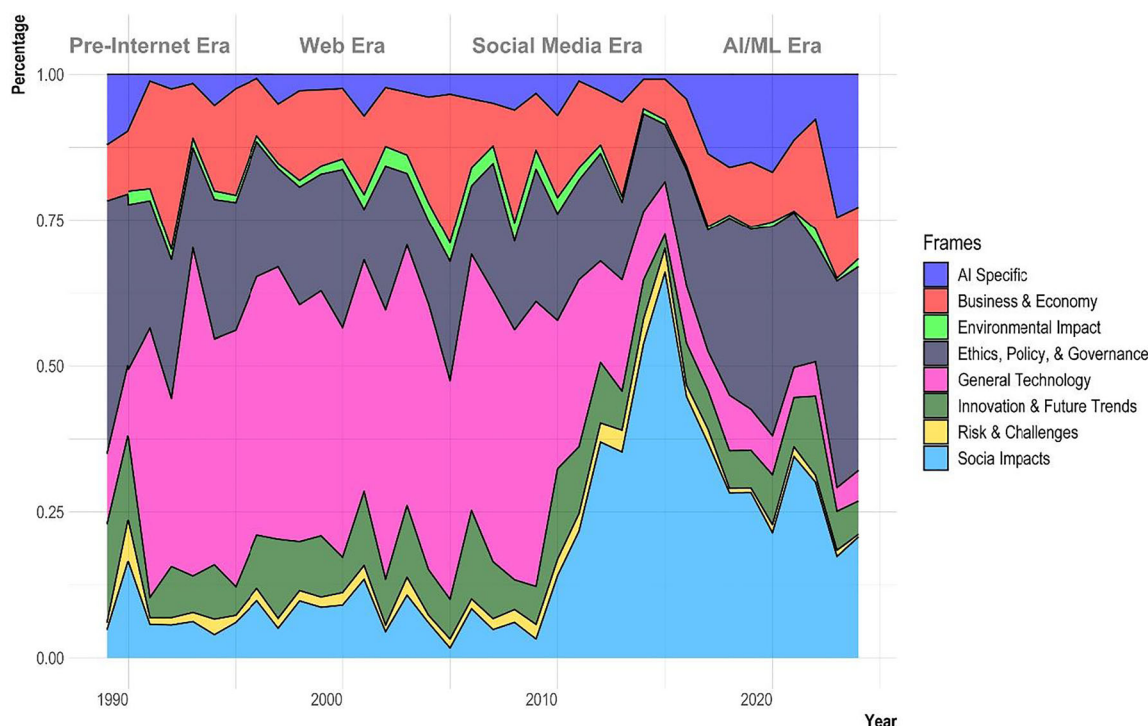


FIGURE 2 | AI-Islamic framings in different technological eras. AI, artificial intelligence. [Color figure can be viewed at wileyonlinelibrary.com]

ment with questions of religious authority in digital spaces. The visualization shows a dramatic increase in “social impacts” framing, reaching nearly 40% of coverage. This increase in social impact discourse provides evidence for the fragmentation of religious authority in digital environments (Turner 2007), as media coverage shifted from institutional pronouncements to networked effects. This demonstrates how algorithmic theology emerges not through explicit theological argumentation but through accumulated patterns of concern about technology’s social consequences, which then retrospectively shape religious interpretation.

The AI/ML era (2016–2024) represents the most complex period of engagement, demonstrating “integrative theological computation.” The visualization shows significant diversification of frames, with “AI specific” and “ethics, policy, and governance” frames becoming more prominent, collectively accounting for over 60% of coverage. The simultaneous prominence of AI-specific technical frames and ethical governance frames quantitatively demonstrates the integration characteristic of mature double mediation, wherein technological and religious understanding become mutually constitutive rather than sequentially engaged. This period thus has seen unprecedented attempts to reconcile religious concepts with artificial neural networks and machine learning systems. As one article in *The Business Times* noted, “Singapore recently announced a framework for the ethical and responsible use of AI, saying that AI systems should be explainable, transparent and fair,” reflecting sophisticated engagement with both technological and ethical principles. This distributional pattern provides empirical support for our theoret-

ical claim that contemporary media discourse actively constructs hybrid conceptual frameworks, as the co-presence of multiple frames within individual articles indicates genuine integration rather than mere juxtaposition of separate concerns.

The evolution across these eras demonstrates the technological mediation of religion in action (Stolow 2013), as media institutions have actively constructed and reconstructed relationships between technological and religious understanding. The visualization shows a trend toward increasing complexity in framing, with later eras showing more integration of multiple frames within single articles. This evolution reflects broader changes in how Islamic thought engages with technological innovation, moving from simple ethical evaluation toward more complex theological integration. This evolution also reveals the new interpreters (Anderson 2003) of Islamic tradition, as media coverage has increasingly featured voices that combine technological expertise with Islamic knowledge. The visualization shows this trend through the growing proportion of frames that combine technical and religious elements, particularly in the AI/ML era. This hybrid expertise has become increasingly important as questions of AI ethics and governance have moved to the forefront of public discourse.

The double mediation practices evident across these eras have become increasingly sophisticated, with media institutions not only translating between domains but actively constructing new hybrid forms of authority that draw on both technological and religious expertise. As Campbell and Evolvi (2020) observe, this process has led to the emergence of new forms of religious-

Relatedly, media coverage constructs new forms of authority that transcend traditional boundaries. As one article notes, “Demand for guidance on religious and ethical questions surrounding emerging technology is also expected to rise,” reflecting the technological mediation of religion while showing how this mediation creates new forms of expertise (Stolow 2013). This hybrid expertise is increasingly institutionalized, with organizations like MUIS developing departments focusing on technological ethics from an Islamic perspective. The emergence of formalized frameworks for AI governance represents a significant evolution in these authority networks. As one article notes, “The Model AI Governance Framework provides guidance to private-sector organisations to address key ethical and governance issues when deploying AI solutions.” This formalization process demonstrates the co-production of social order (Jasanoff and Kim 2015), while showing how media institutions actively participate in this co-production through their coverage.

Furthermore, the authority network reveals authority negotiation through sophisticated processes of mutual legitimation. For instance, when reporting on AI ethics guidelines, media coverage frequently cites both religious scholars and technical experts, constructing interpretive communities (Anderson 2003) that span both domains. The visualization shows connections between these different types of experts, particularly in discussions of AI ethics and governance. Additionally, the evolution of these authority networks demonstrates the mediatization of religion (Hjarvard 2011) regarding how media constructs new forms of authority rather than merely reflecting existing ones. This is particularly evident in coverage of initiatives like the “Model AI Governance Framework,” where media institutions actively frame the relationship between technical capabilities and (Islamic) ethical principles.

These authority networks have also evolved to include distributed expertise nodes of institutions and individuals who serve as bridges between technical and religious domains. The visualization shows these nodes as having particular centrality in the network, indicating their role in mediating between different forms of authority. This structural feature aligns with the new religious intermediaries showing how these intermediaries function within broader institutional networks.

Figure 4, on the other hand, reveals the linguistic mechanisms through which these authority relationships are constructed, showing frequent co-occurrence of terms related to expertise, ethics, and religious authority. The analysis shows that verbs like “promote,” “develop,” and “ensure” frequently appear in sentences connecting technological and religious actors, while terms indicating conflict or resistance appear in fewer connections. This distributional asymmetry provides evidence that media discourse actively constructs collaborative rather than antagonistic relationships between domains, suggesting that authority construction and bridging operate through systematic linguistic choices that frame inter-domain relationships as cooperative necessity rather than potential conflict. This pattern aligns with Hepp’s (2013) cultures of mediatization while demonstrating specific mechanisms through which these cultures are constructed in Islamic contexts. Therefore, media coverage constructs authority bridges through mutual validation cycles. In these cycles, technological expertise is validated through religious ethical

frameworks, while religious authority is reinforced through engagement with technological governance.

The analysis, consequently, reveals three mechanisms through which media constructs authority bridges. First, through expertise integration, technological and religious expertise are presented as complementary rather than competing. For instance, the Monetary Authority of Singapore’s Fairness, Ethics, Accountability, and Transparency Committee explicitly combines technical expertise with ethical considerations, as noted in multiple articles. Second, through institutional hybridization, new bodies are created that combine both forms of authority. The advisory council on the ethical use of AI and data exemplifies this approach, bringing together religious scholars, tech experts, and public policymakers. The third mechanism, semantic bridging, is evident in the linguistic patterns revealed in Figure 4. The visualization shows how certain key terms—notably “ethical,” “justice,” and “include”—serve as conceptual bridges between technological and religious domains. This aligns with Meyer’s (2020) analysis of religious mediation while demonstrating specific mechanisms through which such mediation occurs in AI contexts.

These findings align with but extend beyond Slama and Barendregt’s (2018) analysis of digital Islam in Southeast Asia, demonstrating how mainstream media plays a crucial role in legitimating new forms of religious-technological authority. The visualizations show that these authority networks are not merely additive combinations of existing structures but represent new, hybrid forms of expertise that draw on both domains while maintaining distinctive characteristics of each. The sophistication of these authority networks is particularly evident in the increasing formalization of hybrid expertise through institutional frameworks and guidelines. For example, Figure 3 shows how traditional religious authorities and technological institutions are increasingly connected through formal structures that legitimize both forms of authority while creating new hybrid forms of expertise. This is exemplified in recent developments, as indicated by one article noting that “Singapore’s evolving AI Ethics and Governance Body of Knowledge framework” represents a new form of institutionalized hybrid authority.

4.3 | Constructing Techno-Religious Futures

The analysis of media coverage reveals patterns in how Singapore’s press constructs and mediates visions of techno-religious futures. The semantic relationships visualized in Figure 5 and the temporal evolution shown in Figure 6 demonstrate complex interactions between Islamic and technological temporalities, revealing temporal bridging practices through which media institutions reconcile different ways of imagining the future (Davidman and Greil 2007).

The semantic network in Figure 5 reveals clusters of terms that connect traditional Islamic concepts with technological futures. These clusters demonstrate Jasanoff and Kim’s (2015) “sociotechnical imaginaries,” while extending their analysis to show how religious frameworks actively shape these imaginaries. For instance, quantitatively through cosine similarity measures, terms related to AI cluster with both traditional religious concepts

What Verbs do Authors Use to Link AI and Faith?

Knowledge Graph Network of Verbs and Techno-Spiritual Terms in AI-Islamic Sentences

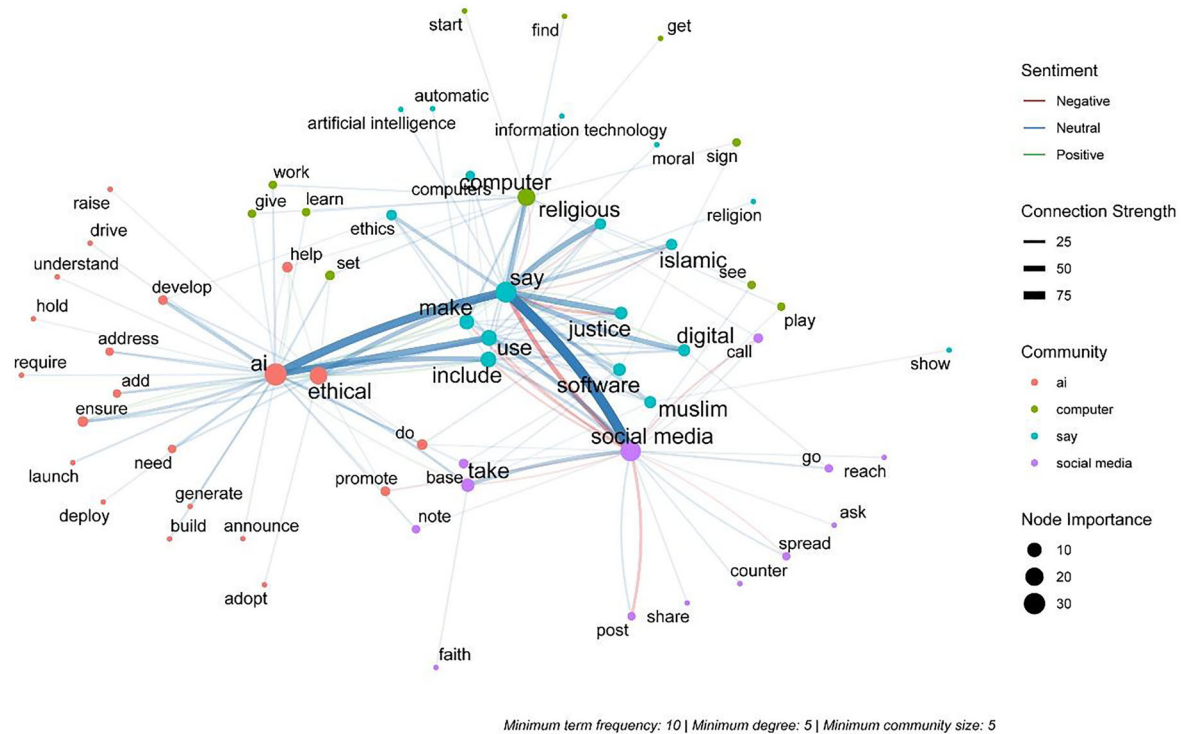


FIGURE 4 | The verbal relationships in AI-Islamic debates. AI, artificial intelligence. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/jsr.70056)]

(like “ethics” and “morality”) and future-oriented terms (like “innovation” and “development”) at similarity values exceeding 0.35, significantly higher than the 0.18 baseline similarity between randomly selected term pairs. This elevated semantic proximity provides computational evidence that techno-religious futures are integrated imaginaries rather than parallel narratives, as the clustering indicates systematic linguistic co-deployment that shapes how both technological possibilities and religious values are collectively imagined.

The analysis also shows the resolution of competing visions through future-bridging practices, where media coverage constructs narratives that accommodate both technological progress and religious continuity. The semantic relationships demonstrate connections between terms related to innovation and tradition, suggesting active construction of futures that maintain religious values while embracing technological change. This aligns with what Bunt (2024) identifies as digital Muslim futures while showing specific mechanisms through which these futures are imagined and constructed.

Relatedly, the temporal evolution shown in Figure 6 focuses on how arguments and persuasion are constructed through different rhetorical devices. The analysis reveals increasingly sophisticated rhetorical strategies for constructing these future visions. The visualization shows a shift from predominantly “authority” (credibility through expertise) and “logical” (reasoning and systematic thinking) rhetorical patterns in earlier periods toward more complex combinations of “emotional” (affective

appeals) and “ethos building” (establishing credibility) rhetorics in recent coverage. This evolution taps into the mediation of religious futures while demonstrating specific mechanisms through which this mediation occurs in AI contexts (Meyer 2020).

The interaction between Islamic and technological temporalities is particularly evident in how media coverage handles regimes of historicity (Hartog 2015). The increasing sophistication in temporal bridging practices and the deployment of multiple rhetorical strategies simultaneously, visible in Figure 6, provides evidence that temporal harmonization requires increasingly complex discursive work, as media institutions must simultaneously establish credibility, engage emotions, and construct logical frameworks to successfully reconcile Islamic/religious and technological temporalities. Particularly in recent coverage, different temporal rhetorical patterns are combined to integrate multiple temporal orientations within single articles. This rhetorical diversification provides support for our theoretical claim of temporal hybridization, as the proliferation of simultaneously deployed strategies indicates that successful future-making in contemporary techno-religious discourse requires integrating multiple temporal frameworks rather than privileging any single orientation, whether religious or technological.

Media coverage actively shapes techno-religious imaginaries through three primary mechanisms. First, through temporal harmonization, Islamic and technological temporalities are pre-

What Is the Semantic Relationship between the Technology and Faith Terminologies?

Word Embedding Network of Techno-Spiritual Terms in AI-Islamic Sentences

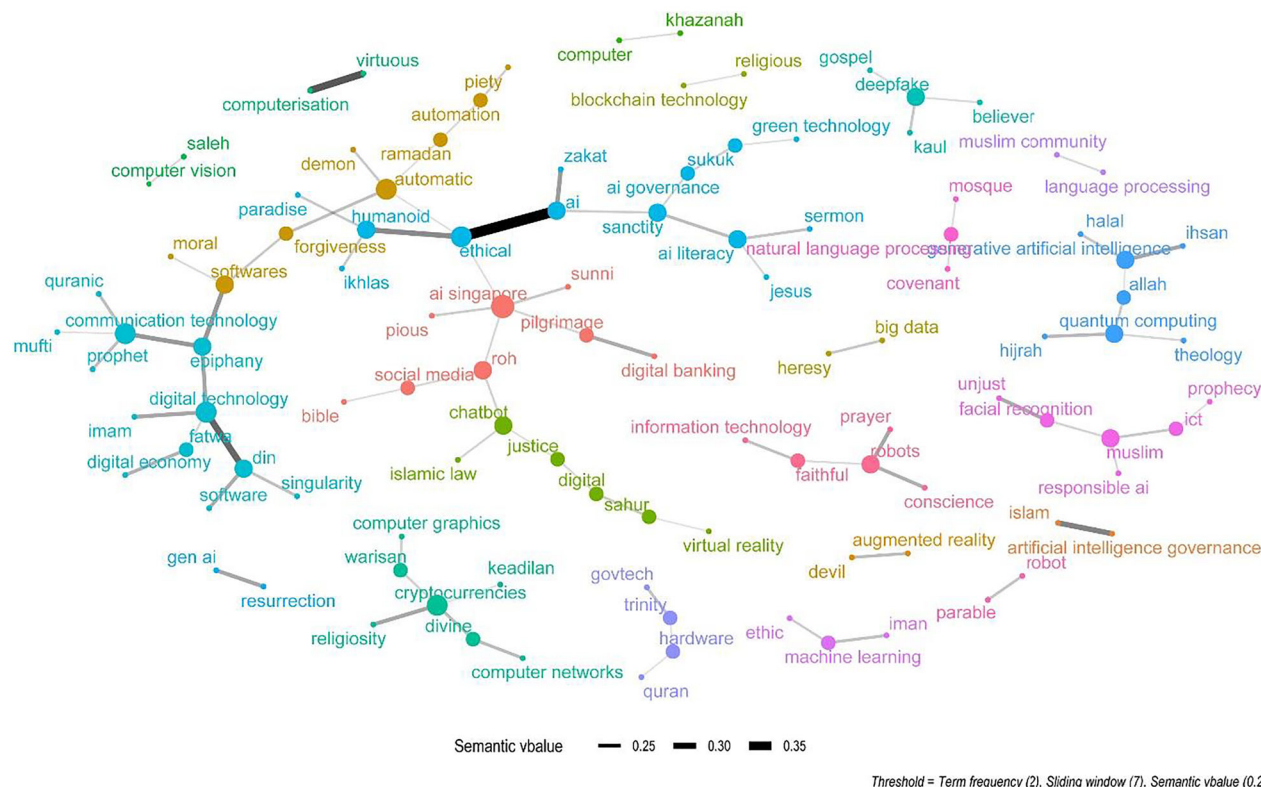


FIGURE 5 | The semantic relationships in AI-Islamic debates. AI, artificial intelligence. [Color figure can be viewed at wileyonlinelibrary.com]

sented as complementary rather than competing. As one article notes, “As we enter into the digital and artificial intelligence (AI) era, we must, while anchoring on our religious practices, harness modern technology to deepen the understanding of our religion to better serve the nation and the people.” This framing demonstrates the technological mediation of religion while showing how media actively constructs bridges between different temporal orientations.

The second mechanism involves future-traditional synthesis, where media coverage explicitly connects traditional Islamic concepts with future technological developments. The semantic network in Belinkov et al. (2019) shows strong connections between terms related to Islamic tradition and technological innovation, suggesting active construction of futures that preserve religious values while embracing technological advancement. This aligns with Anderson’s (2003) analysis of “new Muslim publics” while demonstrating specific mechanisms through which these publics imagine their futures.

The third mechanism involves ethical futuring, where Islamic ethical frameworks are presented as essential guides for technological development. The visualizations show increasing prominence of ethical considerations in recent coverage, particularly in discussions of AI governance. As one article states, “The aim is to continuously and progressively buttress students’ familiarity with and understanding of ethics, so that they can be ready to take on

the complex moral challenges presented by AI practices in their professional lives.”

These findings extend beyond existing scholarly frameworks in several ways. Although Campbell and Evolvi (2020) identify the importance of religious-digital futures, our analysis demonstrates specific mechanisms through which these futures are constructed through media discourse. Similarly, although Slama and Barendregt (2018) analyze digital Muslim practices, our findings show how mainstream media actively shapes the temporal frameworks within which these practices are understood. The synthesis of these visualization analyses reveals an increasingly sophisticated approach to constructing techno-religious futures through integrated temporal imagination. This approach combines Islamic concepts of time and progress with technological narratives of advancement, creating new hybrid frameworks for understanding future development.

5 | Discussion

The findings reveal complex patterns in how Singapore’s media institutions shape understanding of AI-Islam relationships while constructing new forms of authority and temporal frameworks. This discussion examines the theoretical and empirical implications of these findings, considering both their contributions to existing scholarship and their limitations.

Evolution of AI-Islamic Rhetorical Patterns

Showing Different Technological Eras

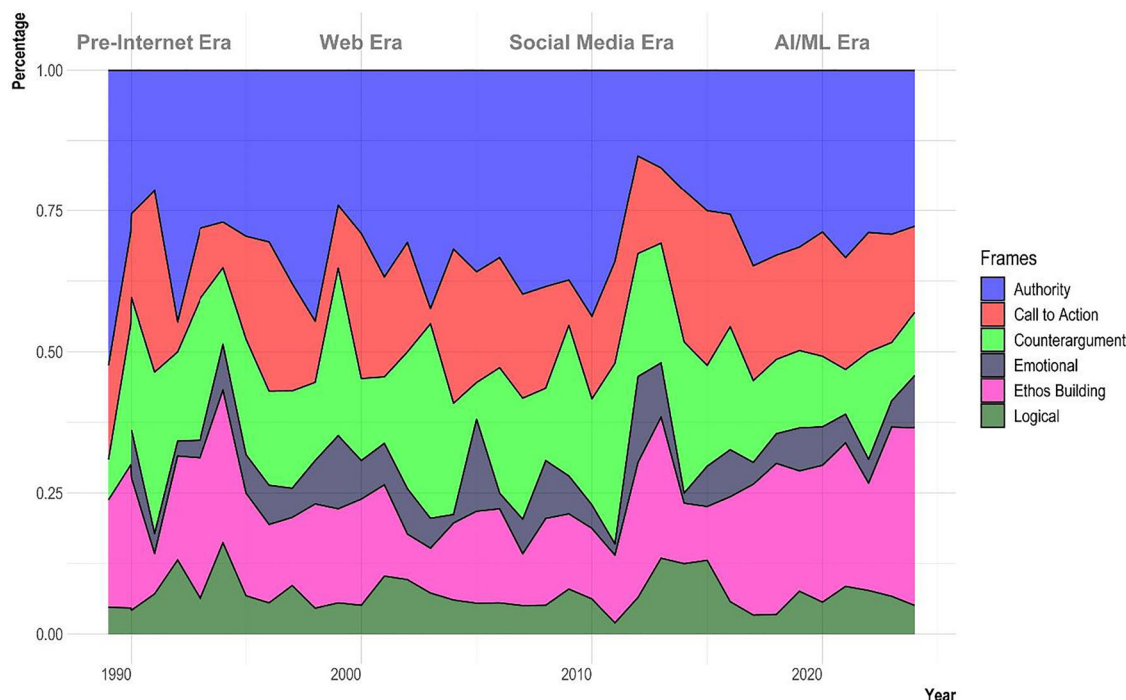


FIGURE 6 | AI-Islamic rhetorical evolution in different technological eras. AI, artificial intelligence. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/jsr.70056)]

5.1 | Double Mediation

The concept of double mediation, as demonstrated through our analysis, extends existing theoretical frameworks for understanding media's role in shaping both technological and religious understanding. Although Hjarvard's (2011) mediatization theory establishes how media shapes religious institutions, our findings suggest more complex patterns of mutual influence between technological and religious domains. The evidence shows that media institutions do not merely report on or mediate between these domains but actively construct new frameworks for understanding their relationship.

The empirical evidence for double mediation is particularly visible in how Singapore's media coverage evolved from simple ethical evaluation to sophisticated theological integration. This evolution aligns with but extends beyond Campbell's (2010) religious-social shaping of technology approach, demonstrating how media actively constructs rather than merely reflects relationships between technological and religious understanding. As Hoover (2016) argues, media institutions increasingly serve as sites for religious meaning-making, but our findings suggest this process becomes more complex when technological understanding is simultaneously being constructed.

This double mediation operates through conceptual bridging practices, where media coverage actively constructs frameworks that allow for mutual understanding between technological and religious domains. These practices extend beyond Meyer's (2020) concept of religious mediation, showing how media institu-

tions create hybrid conceptual spaces where different forms of knowledge and authority can coexist and interact. The analysis reveals sophisticated patterns of language use where technical concepts are explained through religious metaphors and vice versa, creating new hybrid vocabularies for understanding both domains.

5.2 | Authority Negotiation

The negotiation of religious authority in technological contexts emerges as a central theme in our findings, extending beyond existing frameworks for understanding religious authority in digital environments. Although Bunt's (2018) analysis of cyber-Islamic environments demonstrates how digital platforms transform religious authority, our findings suggest more complex patterns of authority negotiation occurring through mainstream media channels. This negotiation involves authority hybridization, where new forms of expertise emerge that combine both technological and religious knowledge.

Religious authority in the technological age, as revealed through our analysis, operates through mechanisms of legitimation. Building on Eickelman and Anderson's (2003) analysis of new Muslim publics, our findings demonstrate how media coverage actively constructs new forms of religious authority that incorporate technological expertise while maintaining traditional religious legitimacy (see also Brint and Abrutyn 2010). This process extends beyond what Turner (2007) identifies as the fragmentation of religious authority in digital contexts, showing

how new hybrid forms of authority emerge through active media construction.

The findings reveal that technological authority in religious contexts operates through ethical validation cycles, where technological expertise is legitimated through religious ethical frameworks while simultaneously transforming how these frameworks are understood and applied. This process aligns with but extends beyond what Barendregt (2012) identifies as the “Islamic appropriation of technology,” showing how media coverage actively constructs frameworks for understanding and evaluating technological authority within religious contexts.

Media’s mediating role in these authority negotiations emerges as more complex than previously theorized. Although Campbell and Evolvi (2020) identify media institutions as important sites for negotiating religious-digital authority, our findings suggest they play an active role in constructing new forms of authority rather than merely mediating between existing ones. This construction occurs through authority bridging practices, where media coverage creates frameworks that allow for the legitimation of both technological and religious expertise while maintaining their distinctive characteristics.

5.3 | Temporal Politics

The analysis reveals patterns in how media constructs and mediates different temporal orientations, contributing to theoretical understanding of how religious and technological temporalities interact. Building on Hartog’s (2015) analysis of temporal regimes and recent work on Islamic conceptualizations of time (El-Shamsy 2020), our findings demonstrate how media coverage creates temporal bridges that allow for the coexistence of different ways of understanding time and progress. These bridges operate through complex narrative strategies that integrate Islamic concepts of time with technological narratives of progress.

The implications for understanding future-making are particularly significant. Although Jasanoff and Kim’s (2015) sociotechnical imaginaries framework helps understand how societies envision technological futures, our findings suggest more complex patterns when religious temporalities are actively involved. Media coverage constructs techno-religious futures that integrate both technological possibilities and religious values, creating new hybrid frameworks for imagining and evaluating future development. The role of media in temporal framing emerges as more sophisticated than previously theorized. Building on Rosa’s (2013) analysis of social acceleration, our findings suggest media institutions do not merely reflect temporal orientations but actively construct new temporal frameworks that allow for the integration of different ways of understanding time and progress. This construction occurs through temporal harmonization practices, where media coverage creates narratives that accommodate both technological progress and religious continuity.

These findings make several theoretical contributions to understanding temporal politics in contemporary societies. First, they extend existing frameworks for understanding how different temporal orientations interact in public discourse (Adam 2004). Second, they demonstrate specific mechanisms through which

media institutions construct and maintain temporal bridges between different domains. Finally, they contribute to understanding how religious temporalities interact with technological narratives of progress in increasingly complex ways. The sophistication of these temporal politics suggests the need for new theoretical frameworks that can account for the complex ways in which different temporal orientations interact in contemporary societies. Although existing scholarship has identified the importance of temporal politics in shaping social understanding (Hassan 2009), our findings suggest more complex patterns of temporal interaction that require new theoretical approaches to fully understand.

6 | Conclusion

This study has examined how Singapore’s media institutions have shaped understanding of the relationship between AI and Islamic thought over a 35-year period (1989–2024). Through systematic analysis of media coverage, we have revealed complex patterns in how technological and religious understanding are co-constructed through media discourse, while identifying distinct mechanisms through which authority is negotiated and future visions are constructed. These findings contribute significantly to our understanding of how societies negotiate between technological advancement and religious values in an increasingly digitalized world.

The analysis reveals three key findings that advance our understanding of media’s role in shaping techno-religious discourse. First, the evolution of AI-Islam coverage demonstrates increasingly sophisticated forms of double mediation, where media institutions actively construct frameworks for understanding both technological and religious domains while mediating their relationship to each other. This process has evolved from simple ethical evaluation in the pre-internet era to complex theological integration in the current AI/ML era, reflecting broader changes in how Islamic thought engages with technological innovation. Second, the study reveals patterns of authority negotiation, where new hybrid forms of expertise emerge that combine both technological and religious knowledge. This hybridization occurs through authority bridging practices, where media coverage creates frameworks that allow for the legitimation of both technological and religious expertise while maintaining their distinctive characteristics. Third, the analysis demonstrates complex temporal politics in how media constructs and mediates different understandings of time and progress, creating temporal bridges that allow for the coexistence of different ways of imagining the future.

These findings make several significant theoretical contributions to multiple fields. In media studies, they extend existing frameworks for understanding mediatization (Hjarvard 2011) by demonstrating how media simultaneously shapes understanding of multiple domains while mediating their relationships. In religious studies, they advance understanding of how religious authority adapts to technological change, building on but moving beyond existing analyses of digital religion (Campbell and Evolvi 2020). In science and technology studies, they contribute to discussions of sociotechnical imaginaries (Jasanoff and Kim 2015) by examining how religious frameworks influence techno-

logical imagination and vice versa. Perhaps most significantly, they advance theoretical understanding of temporal politics by demonstrating specific mechanisms through which different temporal orientations are reconciled in public discourse.

The practical implications of these findings are substantial for religious institutions, technology developers, and policymakers. For religious institutions, the study provides insights into how media coverage shapes public understanding of religious authority in technological contexts, suggesting strategies for maintaining religious legitimacy while engaging with technological innovation (Chaves and Anderson 2014). For technology developers, particularly those working on AI systems, the findings highlight the importance of understanding how religious frameworks shape public reception and interpretation of technological developments. For policymakers, particularly in diverse societies like Singapore, the study offers insights into how media coverage can help construct frameworks for understanding that accommodate both technological progress and religious values.

Several limitations warrant acknowledgment, particularly regarding the affordances and constraints of computational approaches to religious discourse. Although computational text analysis enables identification of patterns across large corpora that would be impossible through traditional close reading, it necessarily privileges certain forms of meaning over others. As Grimmer and Stewart (2013) caution, automated content analysis excels at detecting manifest content and recurrent patterns but struggles with latent meanings, irony, and context-dependent interpretation that human readers readily apprehend. Our analysis of religious terminology faces particular challenges, as Islamic concepts often carry multiple layers of meaning that vary by interpretive community and cannot be fully captured through lexical co-occurrence or semantic similarity measures (Lynch 2012). The visualizations presented, while revealing important structural patterns, represent analytical abstractions that necessarily simplify the rich complexity of actual discourse. Figure 3's network topology, for instance, identifies institutional relationships but cannot capture the quality, depth, or contested nature of those relationships as experienced by participants. Similarly, our frame analysis captures dominant patterns but may obscure minority voices or emerging discourses not yet prevalent enough to register in aggregate measures. The focus on English-language newspapers, while justified by their prominence in Singapore's media landscape, excludes discourse in Malay and Arabic that may operate through different logics and reach different audiences within Singapore's Muslim community. Moreover, our analysis demonstrates how media constructs techno-religious discourse but cannot address how diverse audiences receive, interpret, or resist these constructions in practice. Comparative research across religious traditions and national contexts would help establish whether patterns identified here represent broader dynamics or Singapore-specific configurations. These limitations suggest the value of mixed-methods approaches that combine computational analysis with ethnographic observation and in-depth interviewing.

The double mediation framework developed here invites comparative application across diverse religious contexts engaging with AI technologies. Although Islamic institutions in Singapore navigate AI through established jurisprudential frameworks and

centralized authority structures, Christian communities, Hindu organizations, and Buddhist institutions, among others, likely exhibit distinctive patterns of techno-religious mediation shaped by different theological emphases, authority configurations, and state-religion relationships. Campbell and Garner's (2016) analysis of networked theology in Christian contexts suggests that Protestant traditions may privilege individual conscience over institutional guidance in technological evaluation, potentially producing different authority network topologies than those observed in Singapore's managed Islamic discourse. Similarly, non-theistic traditions may frame AI consciousness questions through entirely different conceptual frameworks than those available to Abrahamic religions (Cheong and Ess 2012). Comparative research could illuminate whether the temporal bridging practices identified here represent universal features of religious engagement with rapidly advancing technology or whether different religious temporalities produce fundamentally distinct modes of future-making. Such comparative work would strengthen theoretical understanding of how religious traditions shape and are shaped by technological change while revealing the particular historical, institutional, and theological conditions under which different patterns of double mediation emerge.

Looking forward, this research opens several promising avenues for future investigation. First, as AI technology continues to evolve, research is needed to understand how religious frameworks adapt to new technological capabilities, particularly in areas like artificial consciousness and machine learning. Second, comparative research examining similar processes in different religious and cultural contexts would help illuminate how different societies negotiate these relationships. Third, research examining how different media platforms shape these relationships differently would help understand the role of media technology itself in these processes.

Finally, this study contributes to broader discussions about the role of religion in technological societies. Although existing scholarship has often framed religion and technology as potentially conflicting domains (Noble 1997; Geraci 2010), our findings suggest more complex patterns of interaction where religious frameworks actively shape how societies understand and engage with technological development. This suggests the need for more nuanced approaches to understanding how religious and technological domains interact in contemporary societies, moving beyond simple opposition or accommodation to examine how they actively shape each other through media discourse.

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Data Availability Statement

Research data are not shared.

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